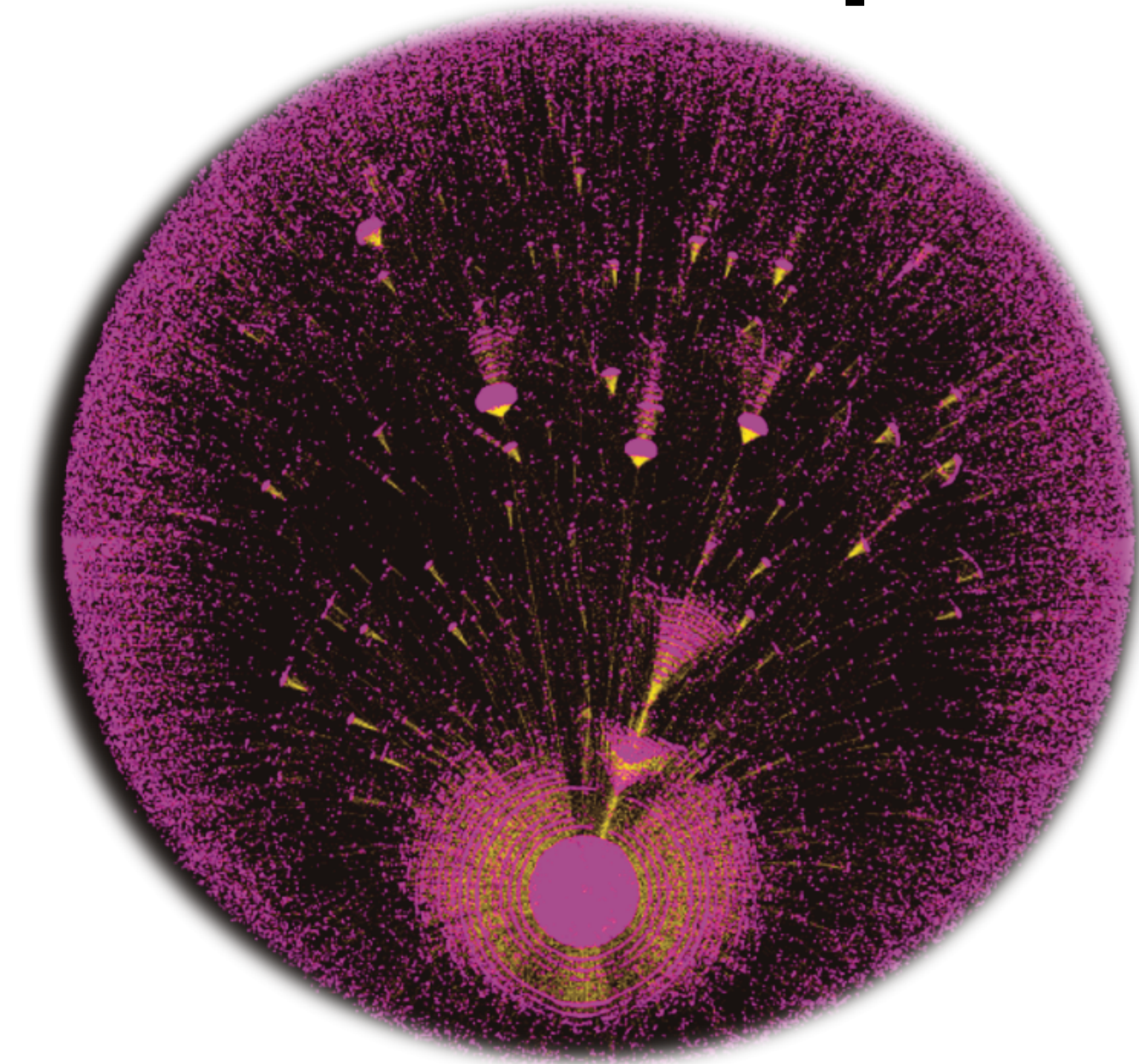


SIR model and link percolation



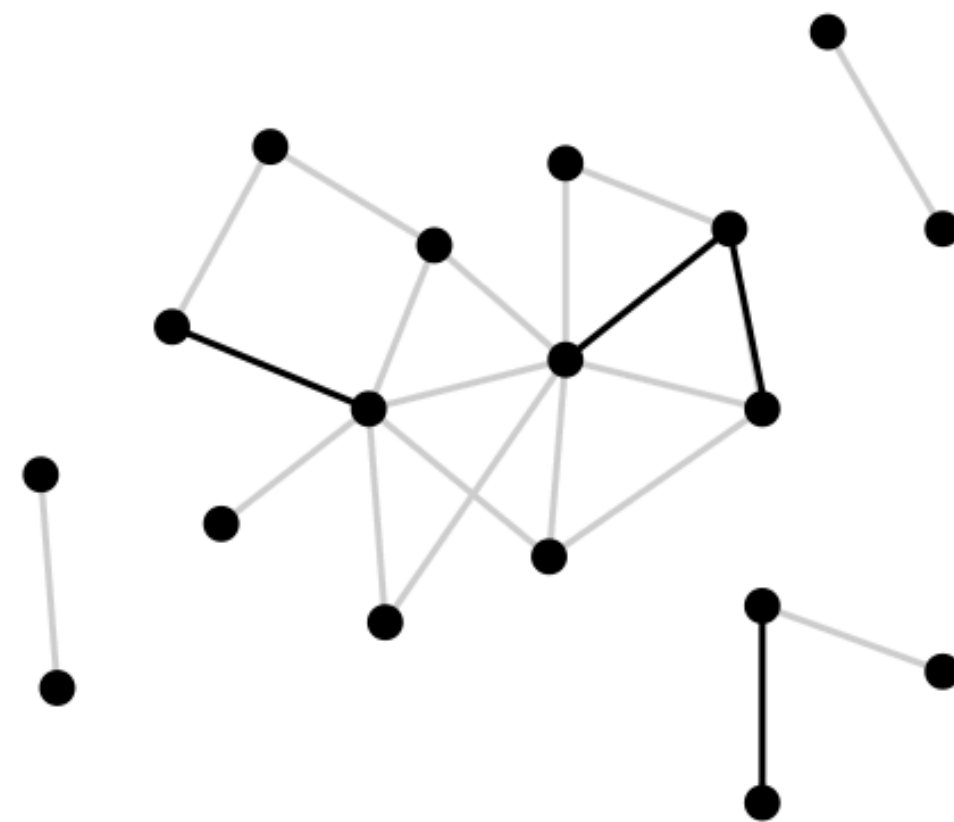
Reference: Chapter 16, Networks, by Mark Newman

Link (bond) percolation - network spreading

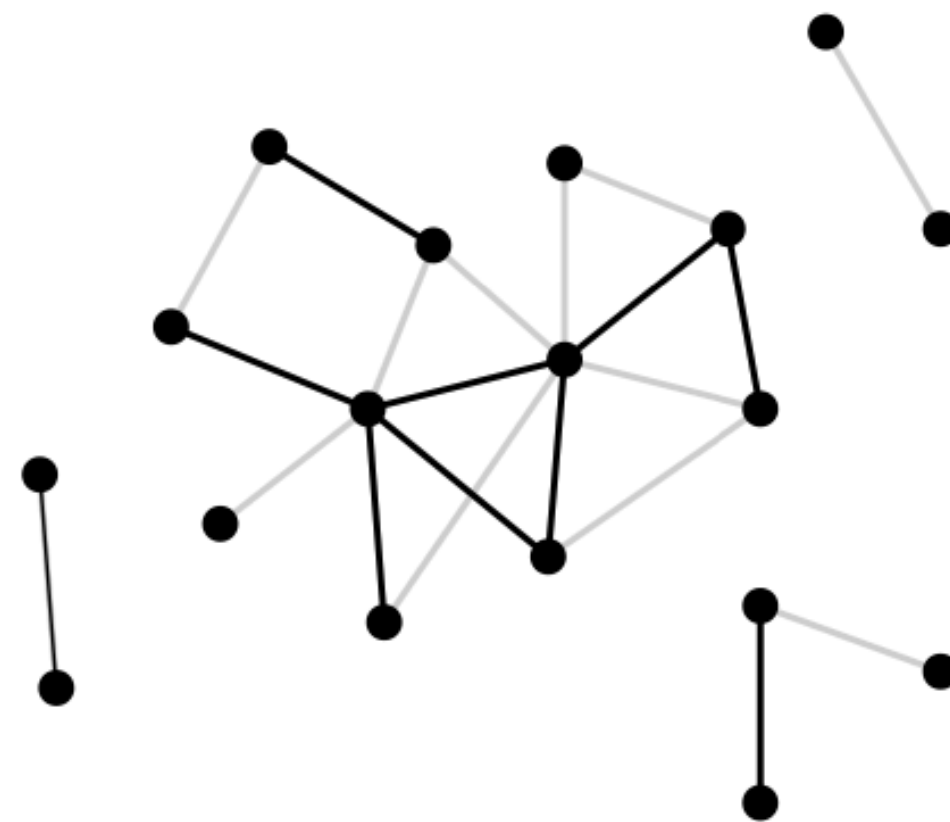
Removal of links from a network, and study the giant cluster behaviour.

Systems	Nodes (spread mechanism)	Giant cluster	Spreading problems
Social network	Friendships (Behavioural influence)	Sharing same behaviour	Obesity contagion
	X/Twitter (Retweet)	Retweeted clusters	Spreading of fake news
	X/Twitter (Retweet)	Retweeted clusters	Maximise outreach like promotion of viral ads
Human contact network	Individuals (Infection)	Infected cluster	If and how much of population will be infected

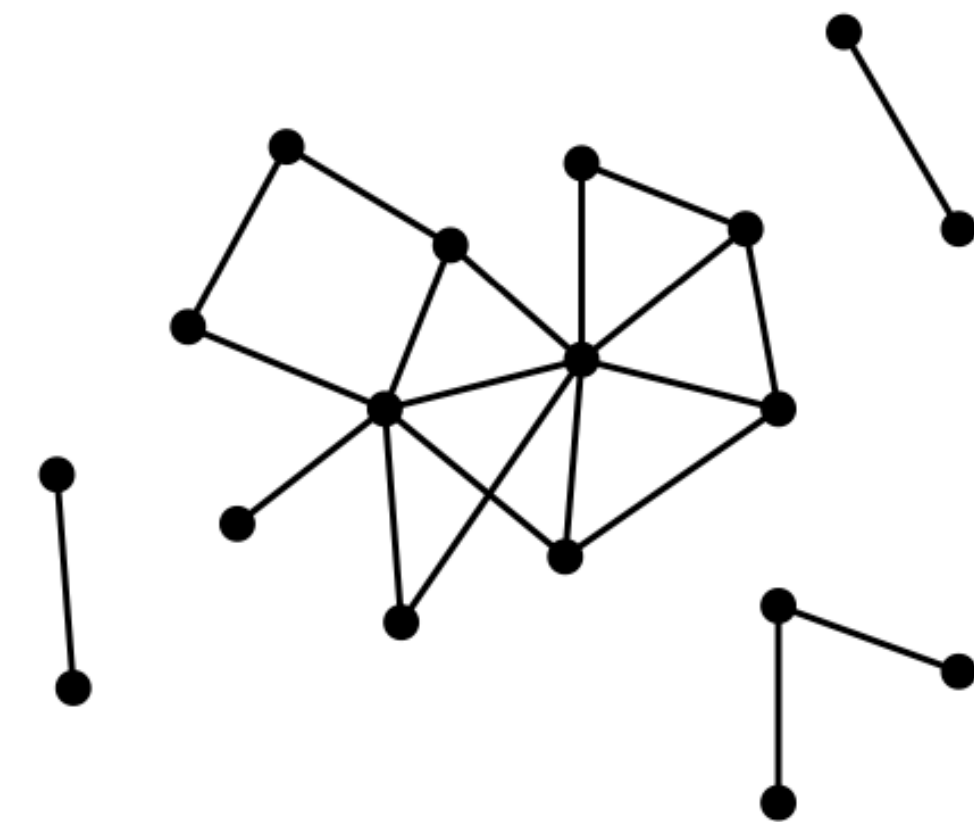
Critical transition of (link) percolation



(a) $\phi = 0.2$



(b) $\phi = 0.5$



(c) $\phi = 1$

Definitions

ϕ

: Probability that a link in the network is occupied

$\phi = 1$ means the original network with all links occupied (spreading occurs over the whole network)

Giant cluster

: Connected component that is finite fraction of the whole network

Percolation

: The state when a giant cluster is present

Quantities of interest:

Giant cluster size: how many nodes are infected S

Percolation threshold: when spread will be pandemic ϕ_c

Random occupation of links

Probability that a random link is occupied: ϕ

Probability that the link is not occupied: $1 - \phi$

Probability that a link does not leads to GC: u

Probability that the link is occupied,
but none of its other k links leads to GC: ϕu^k

If the link leads to a node with excess degree k , probability that this link does not lead to GC: $1 - \phi + \phi u^k$

u is also the sum over all possible k excess degrees:

$g_1(z) = \sum_{k=0}^{\infty} q_k z^k$ is the GF of excess degrees

$$u = \sum_{k=0}^{\infty} q_k (1 - \phi + \phi u^k) = 1 - \phi + \phi \sum_{k=0}^{\infty} q_k u^k = 1 - \phi + \phi g_1(u),$$

Same equation as node percolation!
Meaning same critical threshold ϕ_c

Probability that a random node of degree k
is not connected to GC:

$$\sum_k p_k u^k = g_0(u)$$

$$\phi_c = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle} = \frac{1}{\langle k \rangle_{\text{excess}}}$$

Probability that a random node is in GC:
This is also the size of GC

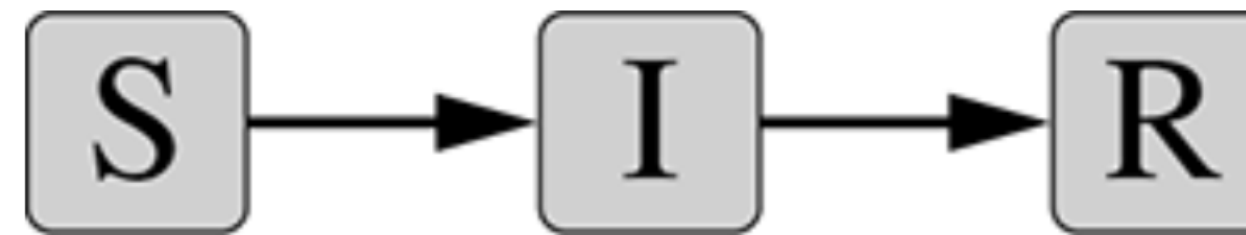
$$S = 1 - g_0(u)$$

Slightly different equation as node percolation!

Disease on scale free networks
always widespread as $\phi_c \approx 0$

SIR (Susceptible-Infected-Recovered) model

An infectious disease (typically virus) causes a person to transition among 3 states:



Susceptible: Not infected yet

Infected: Infected with the disease and being infectious to others

Recovered: Recovered after infection and will never be infected again (immunity)

Sometimes R is also called 'removed' since it is effectively removed from the network.

Fully mixed model

$$\frac{ds}{dt} = -\beta sx,$$

$$\frac{dx}{dt} = \beta sx - \gamma x,$$

$$\frac{dr}{dt} = \gamma x,$$

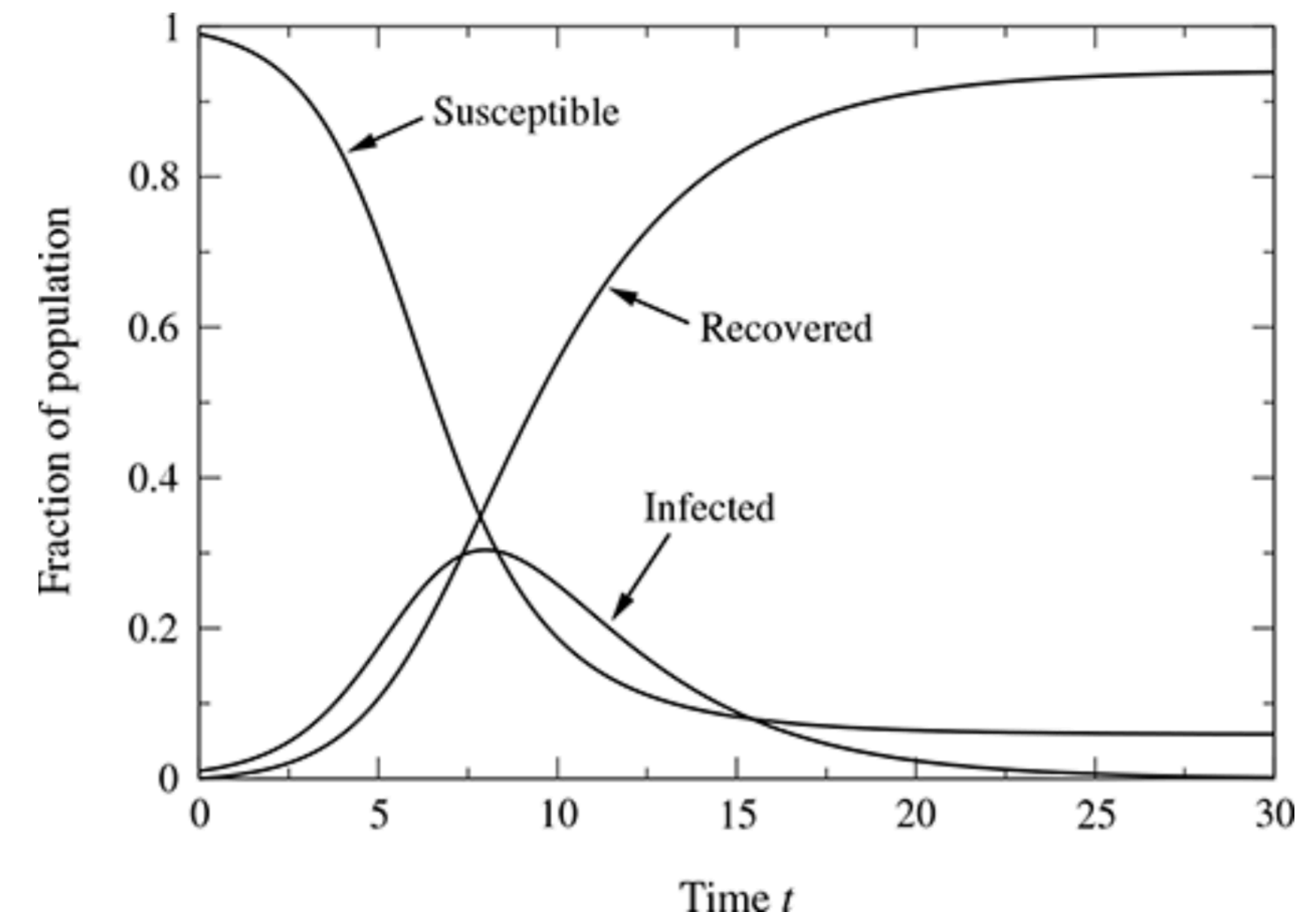
s : fraction of susceptible individuals

β : infection rate

x : fraction of infected individuals

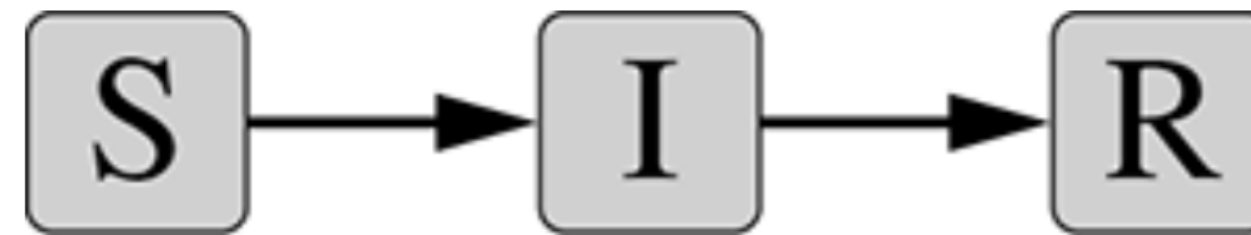
γ : recovery rate

r : fraction of recovered individuals



SIR model for information spreading

A virus - a particular tweet on X/Twitter



Susceptible: account that has not retweeted this tweet

β : chance to retweet by a follower (infection rate)

Infected: account that has retweeted this tweet from a *followee* and remains visible to its followers

γ : rate that the retweet loses visibility by a follower (recovery rate)

Recovered: Account that has retweeted but no longer visible to followers due to its new tweets

Chance of not recovered at time τ :

$$\lim_{\delta\tau \rightarrow 0} (1 - \gamma\delta\tau)^{\tau/\delta\tau} = e^{-\gamma\tau}$$

Chance of infecting a neighbor over time τ :

$$s(\tau) = \beta\tau$$

Chance of recovery between time τ and $\tau+d\tau$: $\gamma d\tau$

Chance of infecting a neighbor eventually at $\tau = \infty$

$$\phi = \int_0^{\infty} r(\tau)s(\tau) = \beta\gamma \int_0^{\infty} \tau e^{-\gamma\tau} d\tau = \frac{\beta}{\gamma}$$

Chance of recovery exactly at time τ :

$$r(\tau) = e^{-\gamma\tau} \gamma d\tau$$

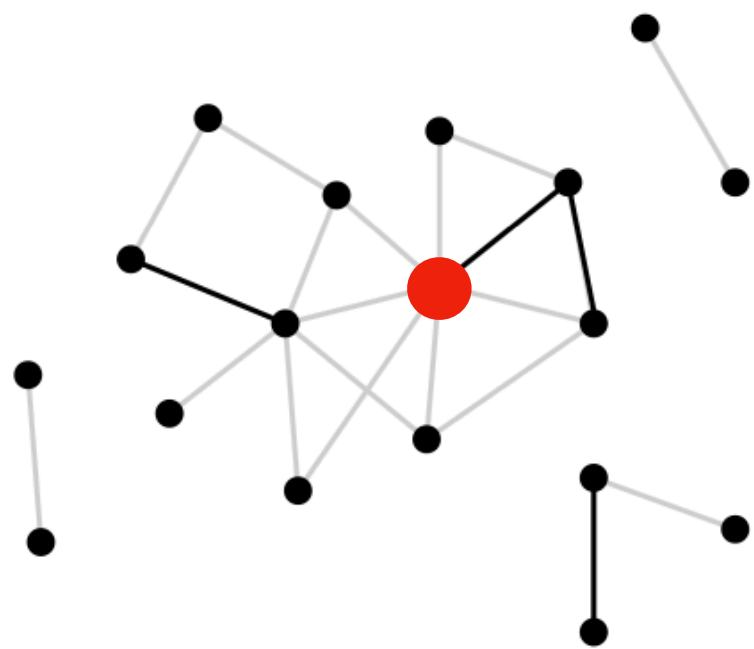
SIR and link percolation equivalence

Chance of retweet by a follower

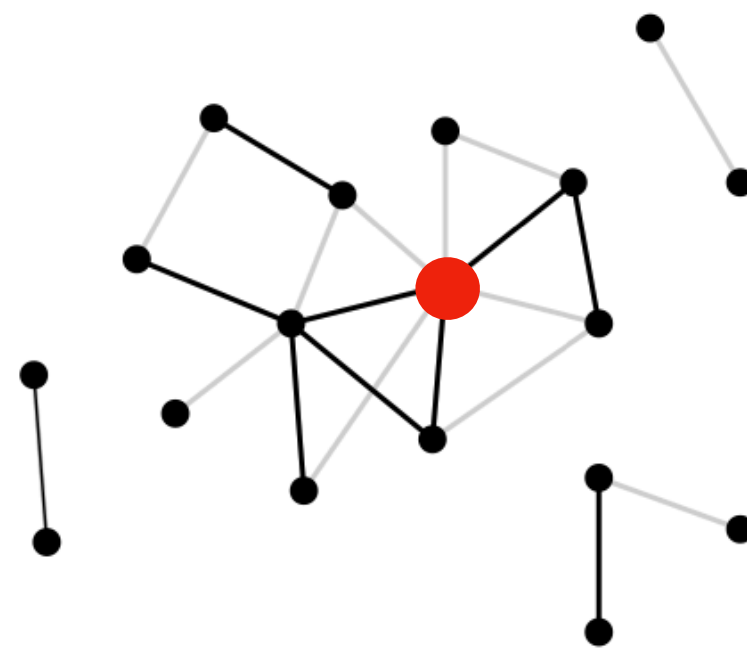
$$\phi = \frac{\beta}{\gamma}$$

Chance of the follower link being occupied

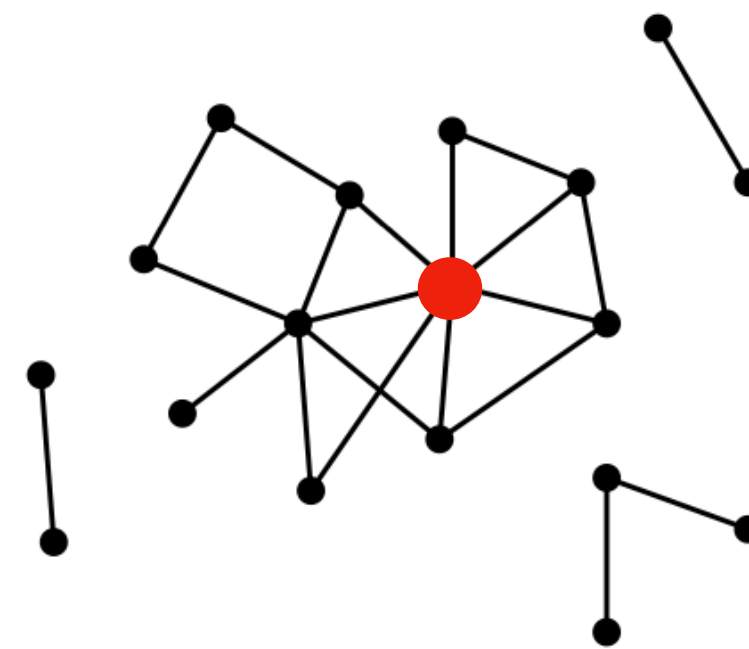
The cluster of connected nodes containing the original 'tweeting account' is the same as the people who has retweeted the tweet. The same picture applies for the virus: it is the people who have been infected the disease (and eventually recovered) when spreading stops.



(a) $\phi = 0.2$



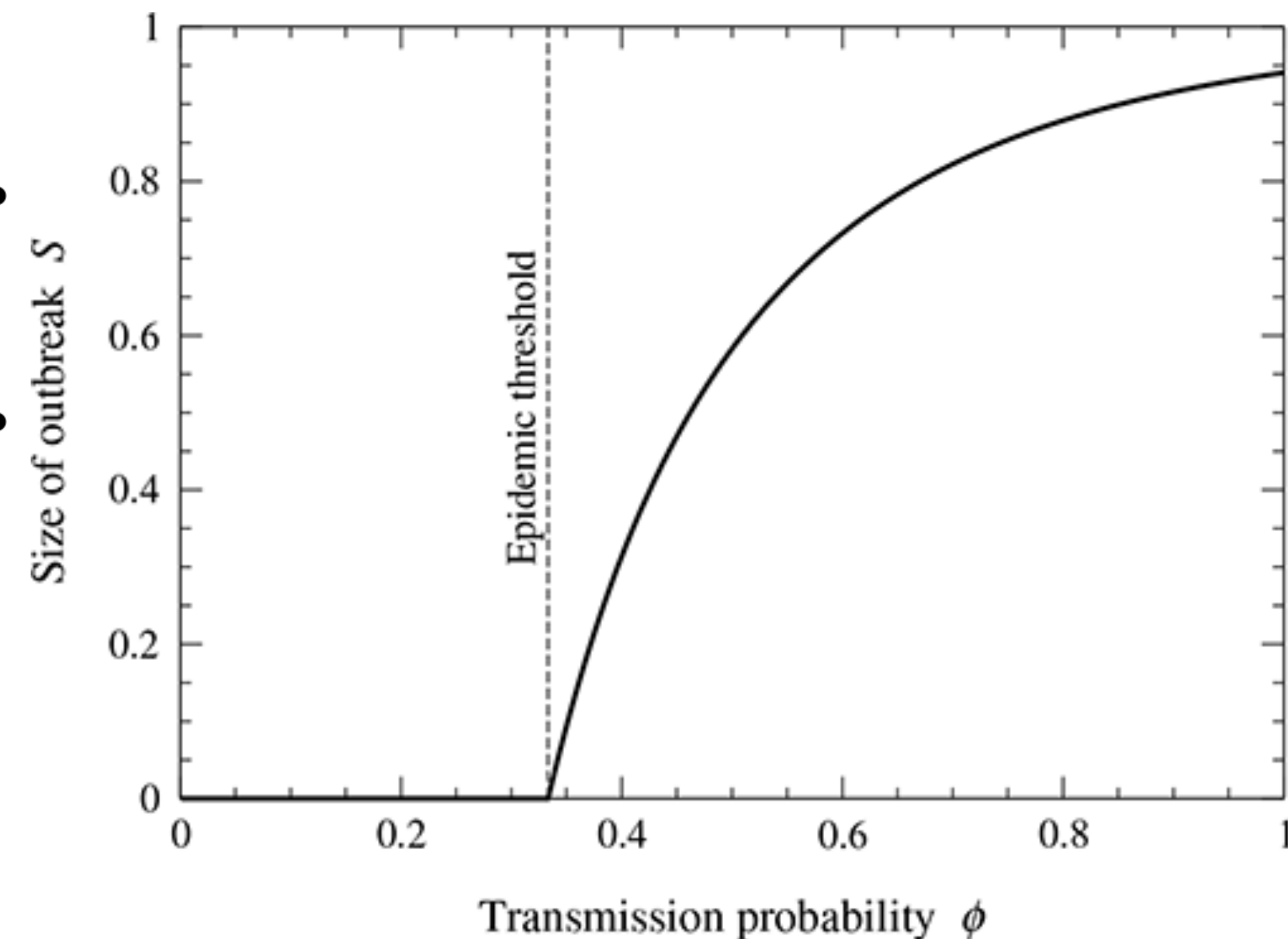
(b) $\phi = 0.5$



(c) $\phi = 1$

Epidemic threshold = link percolation threshold

Epidemic size = giant cluster size



Fully mixed vs. on networks

Fully mixed model

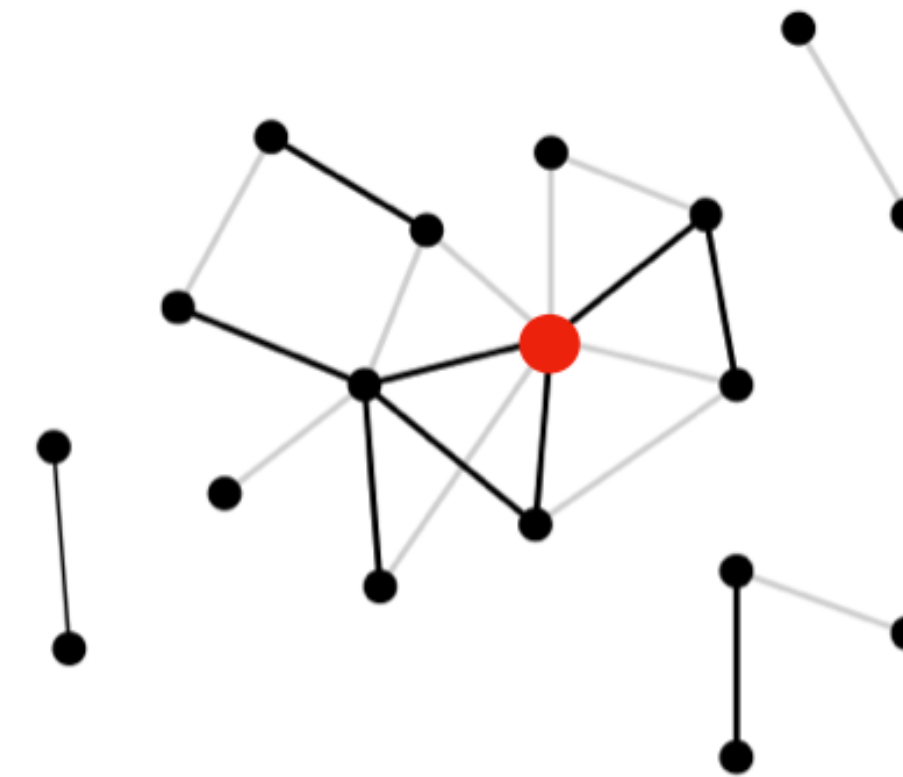
$$\begin{aligned}\frac{ds}{dt} &= -\beta s x, \\ \frac{dx}{dt} &= \beta s x - \gamma x, \\ \frac{dr}{dt} &= \gamma x,\end{aligned}$$

Transition (epidemics) is defined as the R_0 value,
or the *basic reproduction number*

$$R_0^c = \frac{\beta}{\gamma} = 1$$

It is equivalent to fully connected network

Network model

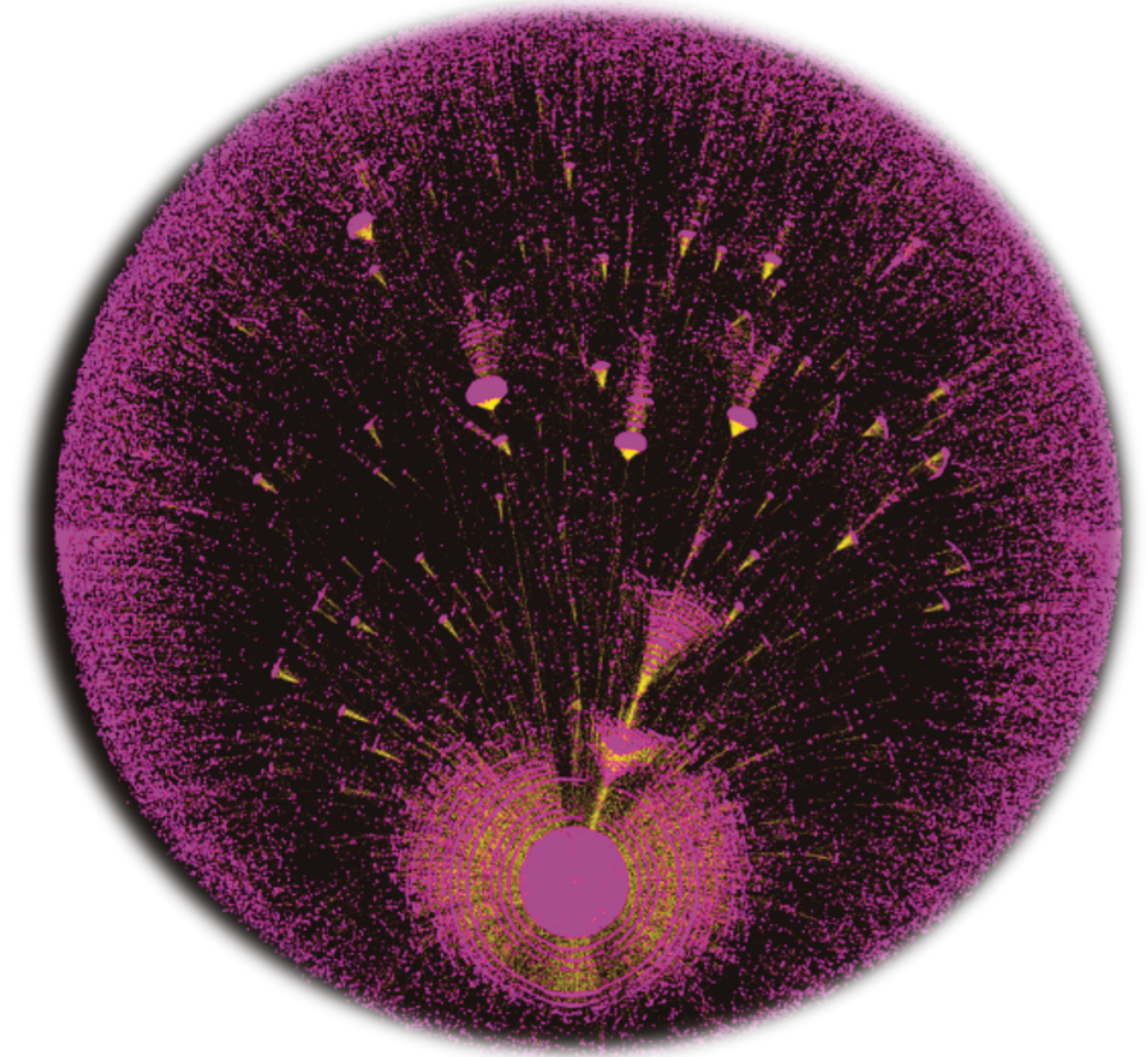


Transition (epidemics) is defined as the ϕ_c value.

$$\frac{\beta}{\gamma} = \phi_c = \frac{1}{\langle k \rangle_{excess}}$$

It depends on average excess degree instead!
More importantly, it depends on network structure.

Viral marketing (spreading)



Organisational Power of Online Social networks



- 2011 London Riots
- 3443 crimes within days
- 3100 people arrested
- \$300 million damage
- Blackberry Messenger???



Arab Spring

Organized through Social media
Facebook
Twitter



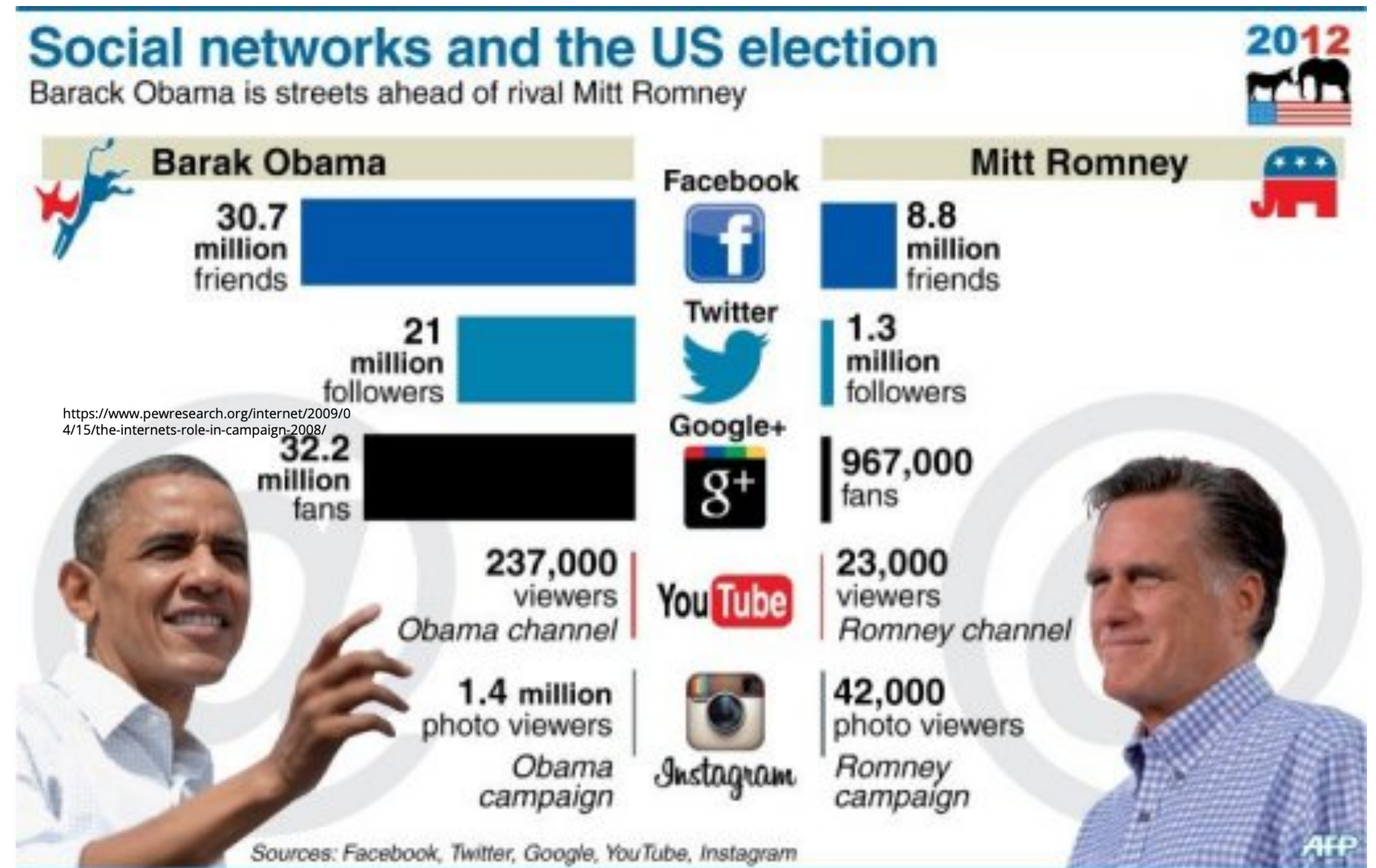
Promotional power of social networks

YouTube



2004, viral video uploads by fans

“Look at you, you filthy downloaders!”



2008 US election, 1st to use social networks

<https://phys.org/news/2012-10-obama-romney-social-media-face-off.html>
<https://www.pewresearch.org/internet/2009/04/15/the-internets-role-in-campaign-2008/>